Overheads of Hardware vs. Software Virtualization

A quantitive analysis of KVM and QEMU TCG Memory subsystems

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Background

- Virtualization is widely-used in cloud computing, OS development. etc.
- Allows you to run multiple *guest operating systems* inside a *host operating system*.

- Most common methods of virtualization
  - **Hardware-Assisted**: Powered by KVM in the Linux kernel for virtualization that runs on the same architecture as the host. Most of the processing is done on the actual hardware, alongside the host! By the help of CPU Virtualization technologies.
  - **Emulation:** For running OSes that run on entirely different CPU architectures! Powered by QEMU's TCG (Tiny Code Generator).

Motivation & Problem

introduce the challenge of virtual memory and two approaches of ept and softmmu introduce the challenge of translation of instructions and how tcg is different from kvm. (Not very complex)

Goal

Explain the goal of trying to measure differences via tools like perf and benchmarks

System Setup

Explain the cpu isolation (hyper threading disabled), running debian nocloud images via scripts and communicating with them via ssh and perf. one is arm and one is x86

Benchmarks

Explain the benchmarks that we have with brief explanation and what stress do they put.

Benchmark 1 results and analysis

Benchmark 2 results and analysis

Benchmark 3 results and analysis

Benchmark 4 results and analysis